

FIT3152 Data analytics – 2026: Assignment 1

Your task	- Analyse the country level predictors of confidence in social organisations and how these change over time using data from the World Values Survey. - This is an individual assignment.
Value	- This assignment is worth 25% of your total marks for the unit. - It has 40 marks in total.
Suggested Length	8 – 10 A4 pages, approximately 2,000 words (for your report) + extra pages as appendix for your R script and report on how Generative AI used, if required. - Font size 11 or 12pt, single spacing.
Due Date	11.55pm Friday 17th April 2026
Submission	- Submit a single PDF file and single video file on Moodle. - Note that submission of a video report is a <u>hurdle requirement</u>. - Use the naming convention: <i>FirstnameSecondnameID.{pdf, mp4, mov etc.}</i> - Turnitin will be used for similarity checking of all written submissions.
Generative AI Use	In this assessment, you can use generative artificial intelligence (AI) in order to <u>search for R functions and examples to perform tasks that you specify</u> only. Any use of generative AI must be appropriately acknowledged (<u>see Learn HQ</u>).
Late Penalties	5% (2 mark) deduction per calendar day for up to one week. - Submissions more than 7 calendar days after the due date will receive a mark of zero (0) and no assessment feedback will be provided.

Instructions

Address each of the research questions below and report the results of your analysis and your interpretation of those results. Report any assumptions you've made in your analysis. Include your R code as an appendix. Your R code must be machine readable text as the university requires all student submissions to be processed by plagiarism detection software. See information on Generative AI below.

There are two options for compiling your written report:

- (1) You can create your report using any word processor with your R code pasted in as machine-readable text as an appendix, and save as a pdf, or
- (2) As an R Markdown document that contains the R code with results and discussion interleaved. Render this as a HTML file and save as a pdf.

Your video report should be less than 100MB in size. You may need to reduce the resolution of your original recording to achieve this. Use a standard file format such as .mp4, or mov for submission.

Software

It is expected that you will use R for your data analysis, graphics and tables. You are free to use any R packages you need but must document these in your report and include in your R code.

Use of Generative AI

AI & Generative AI tools may be used in GUIDED ways within this assessment/task as per the guidelines provided.

In this assessment, you can use generative artificial intelligence (AI) in order to search for R functions and examples to perform tasks that you specify only. Any use of generative AI must be appropriately acknowledged (see Learn HQ).

If you do use Generative AI for your assignment, then you must include the statement "Generative AI was used in this assignment." in the introductory/first paragraph of your report. You must also include the following information as an appendix in your report: (1) the technology you used (e.g. ChatGPT), (2) the information that was generated (e.g. R code fragments), (3) the prompts used (i.e. the questions you asked), and (4) how the output was used in your work.

If you did not use generative AI in your assignment, then include the statement "Generative AI was not used in this assignment." in the introductory/first paragraph of your report.

Questions

The World Values Survey (WVS) is an international research program that studies the social, political, economic, religious and cultural attitudes and values of people around the world. You can read more here: <https://www.worldvaluessurvey.org/WVSContents.jsp>.

For this assignment you will analyse data collected over Waves 1 - 7, from 1981 to 2022. The aim of this assignment is to understand country-level differences in participant responses and the predictors of confidence in social organisations, and how these responses and predictors of confidence have changed over time.

Social organisations include aspects of society such as religion, armed forces, the press, television, trade unions, police, the courts, government, banks, and international and environmental organisations etc. They are indicated in your data by column names having the prefix "C". Predictor variables (**attributes**) include personal information such as age and gender, happiness indicators, attitudes and values towards others, political and social views and participation.

Each student will be assigned a **different** subset of organisations and attributes to study. Your task is to analyse **all** the survey data assigned to you, with a **focus** on the country you have been allocated.

1. Descriptive analysis. (5 Marks)

(a) Describe the data overall, including things such as dimension, data types, distribution of numerical responses, variety of non-numerical (text) responses, missing values, and anything else of interest or relevance.

2. Focus country vs all other countries as a group (independent of time). (13 Marks)

For Question 2 ignore the effect of time. That is, do not separate your data by years or waves when answering the questions below.

(a) Identify your focus country from the accompanying list (**WVSFocusCountry.pdf**). How do participant responses for your focus country differ from the other countries in the survey (treating them as a group)?

(b) How well do participant responses (attributes) predict confidence in social organisations in your **focus country**? Which attributes seem to be the best predictors? Confidence in which social organisations can be more reliably predicted? Explain your reasoning.

(c) Repeat Question 2(b) for the **other countries** as a group. Which attributes are the strongest predictors? Confidence in which social organisations can be more reliably predicted? How do these results compare to those of your focus country?

3. Focus country vs all other countries as a group (over time). (12 Marks)

For Question 3 study the effect of time by separating your data by years or waves when answering the questions below.

(a) How do participant responses for your **focus country** vary over time (using either years or successive waves)? Describe these changes over time and comment on whether they are significant or not. Perform the same analysis for the **other countries** (as a group) and compare the results with your focus country. What are the major differences between the two groups? Create a graphic enabling a reader to compare results (focus vs other countries) over time, for the most interesting results. Describe your reasoning for the design of the graphic.

(b) How does the ability of participant responses (attributes) to predict confidence in social organisations in your **focus country** change over time? Do the important attributes for predicting confidence change over time? Perform the same analysis for the **other countries** (as a group) and compare the results. What are the major differences between the two groups? Create a graphic enabling a reader to compare results (focus vs other countries) over time, for the strongest predictors. Describe your reasoning for the design of the graphic.

4. Video Presentation: (Submission Hurdle and 4 Marks)

Record a short presentation using your smartphone, Zoom, or similar method. Your presentation should be approximately 5 minutes in length and summarise your main findings for Sections 1 – 3, as well as describing how you conducted your research, any assumptions made, and how you designed your graphics.

5 Overall considerations (6 Marks)

This includes: the quality and clarity of your reasoning and assumptions; the strength of support for your findings; the quality of your writing in general and communication of results; the quality of your graphics throughout; the quality of your R coding.

Data

The data for this assignment is a reduced version of the World Values Survey Waves 1 -7 data. The filename is "WVSEExtract.csv". The data includes ordinal data coded on a numerical scale. For this assignment assume it is reasonable to treat these responses as numerical.

Create your individual data as follows:

```
rm(list = ls())
set.seed(12345678) # Your Student Number
VCData = read.csv("WVSEExtract.csv")
VC = VCData[sample(1:nrow(VCData),100000, replace=FALSE),]
VC = VC[,c(1:3,sort(sample(4:50,25,replace=FALSE)),
sort(sample(51:65,8,replace=FALSE)))]
#write.csv(VC, "FIT3152A1Data_YourName.csv", row.names = FALSE)
```

You can save the "VC" file you created by uncommenting the last line above. You can then delete the WVSEExtract.csv file you downloaded.

Locate your focus country using the accompanying document WVSFocusCountry.pdf. A list matching country names with three letter code is in WVSCountryCodes.pdf.

Data fields and brief descriptor

Most fields are on integer scales over varying range. The convention is that larger numbers generally indicate greater agreement with statement or frequency of occurrence. Some exceptions given below. Fields in bold indicate confidence in social organisations.

You can access more detail on each field in your data from the *WVS-7 Master Questionnaire 2017-2020 English.pdf*, linked from <https://www.worldvaluessurvey.org/WVSDocumentationWV7.jsp>. Use the question ID given in the **WVS Wave 7 Reference** in the table below.

Column Name	Original Descriptor	WVS Wave 7 Reference
Wave	Chronology of EVS-WVS waves	A_WAVE
Country	ISO 3166-1 alpha-3 country code	B_COUNTRY_ALPHA
Year	Year survey	A_YEAR
ILFam	Important in life: Family	Q1
ILFriends	Important in life: Friends	Q2
ILLeisure	Important in life: Leisure time	Q3
ILPolitics	Important in life: Politics	Q4
ILWork	Important in life: Work	Q5
ILReligion	Important in life: Religion	Q6
Happy	Feeling of happiness	Q46

Health	State of health (subjective)	Q47
ICQIndependence	Important child qualities: independence	Q8
ICQHardWork	Important child qualities: Hard work	Q9
ICQResonsibility	Important child qualities: feeling of responsibility	Q10
ICQImagination	Important child qualities: imagination	Q11
ICQTolerance	Important child qualities: tolerance and respect for other people	Q12
ICQThrift	Important child qualities: thrift saving money and things	Q13
ICQDetermination	Important child qualities: determination perseverance	Q14
ICQFaith	Important child qualities: religious faith	Q15
ICQUnselfishness	Important child qualities: unselfishness	Q16
ICQObedience	Important child qualities: obedience	Q17
ACTReligion	Active/Inactive membership of church or religious organization	Q94
ACTSport	Active/Inactive membership of sport or recreation	Q95
ACTArtsEd	Active/Inactive membership of art, music, educational	Q96
ACTUnions	Active/Inactive membership of labour unions	Q97
ACTPolParty	Active/Inactive membership of political party	Q98
ACTEnvOrg	Active/Inactive membership of environmental organization	Q99
ACTProfessional	Active/Inactive membership of professional organization	Q100
ACTHumanitarian	Active/Inactive membership of charitable/humanitarian organization	Q101
Happy	Feeling of happiness	Q46
Health	State of health (subjective)	Q47
Trusted	Most people can be trusted	Q57
LifeSatis	Satisfaction with your life	Q49
FutureWork	Future changes: Less importance placed on work	Q43
FutureTech	Future changes: More emphasis on technology	Q44
FutureRespect	Future changes: Greater respect for authority	Q45
PolInterest	Interest in politics	Q199
PolPetition	Political action: Signing a petition	Q209
PolBoycott	Political action: joining in boycotts	Q210
PolDemons	Political action: attending lawful/peaceful demonstrations	Q211
PolScale	Self positioning in political scale	Q240
IncomeEquality	Income equality	Q106
PrivateState	Private vs state ownership of business	Q107
GovtRespons	Government responsibility	Q108
Competition	Competition good or harmful	Q109
HardWork	Hard work brings success	Q110
PolLeader	Political system: Having a strong leader	Q235
PolExperts	Political system: Having experts make decisions	Q236
PolArmy	Political system: Having the army rule	Q237
PolDemoc	Political system: Having a democratic political system	Q238
Sex	Sex	Q260
Age	Age	Q262
CChurches	Confidence: Churches	Q64

CArmedForces	Confidence: Armed Forces	Q65
CPress	Confidence: The Press	Q66
CUnions	Confidence: Labour Unions	Q68
CPolice	Confidence: The Police	Q69
CParliament	Confidence: Parliament	Q73
CCivilService	Confidence: The Civil Services	Q74
CTelevision	Confidence: Television	Q67
CGovernment	Confidence: The Government	Q71
CPolParties	Confidence: The Political Parties	Q72
CMajComp	Confidence: Major Companies	Q77
CEnvProt	Confidence: The Environmental Protection Movement	Q79
CWomensMvt	Confidence: The Womens' Movement	Q80
CCourts	Confidence: Justice System/Courts	Q70
CEU	Confidence: The European Union	Q82_EU